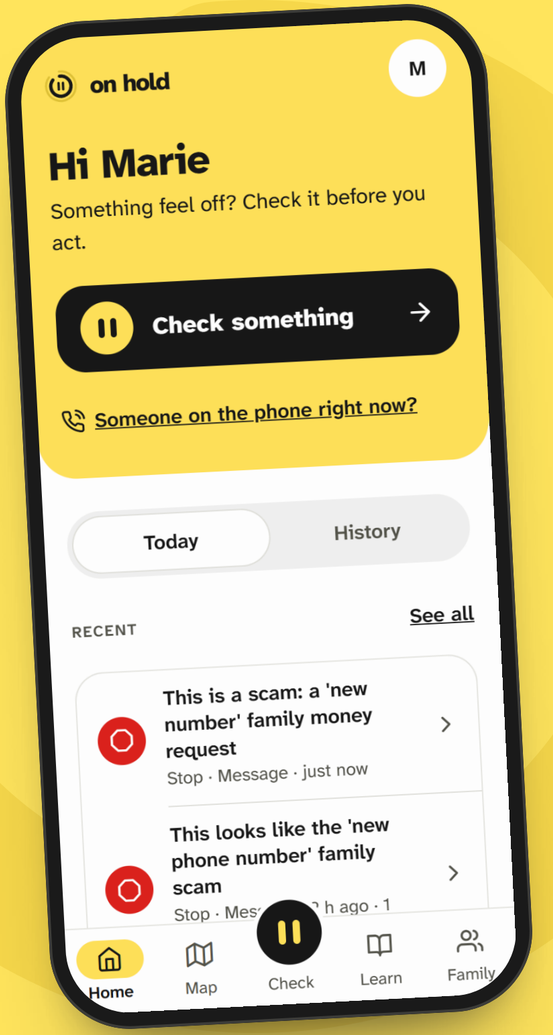
on hold

Pause before you pay.

An AI scam checker for older South Africans.



Victor du Preez · 241250

DV300 Theme 4: AI-Driven Application · Social impact

THE PROBLEM

R2.4 billion

lost to digital banking crime in South Africa in 2025.

70%+

of those losses came through banking apps.

> **“Most digital banking crime was driven by social engineering and the manipulation or compromise of customers.”**
>
> SABRIC

The weak point isn’t the app. It’s the moment someone is talked into acting.

Sources: SABRIC 2025 Annual Banking Crime Statistics (sabric.co.za); SABRIC as quoted by BusinessTech.

WHERE SCAMS WIN

Every scam needs one moment: you acting before you check.

PHONE CALL

“This is the fraud department. Read me the code we just sent so we can stop the payment.”

SMS

“SASSA: Your SRD R370 payment is ON HOLD. Update your details within 24hrs.”

WHATSAPP

“Hi Ma, it’s me. This is my new number. Can you send R2,500 urgently?”

SMS

“AARTO: Outstanding fine R500. Pay within 24hrs for 50% discount.”

Fear **Authority** **Love** **A bargain** and always: hurry.

WHO IT'S FOR

Built for the person being targeted, and the family who looks out for them.

PROTECTED

Older South Africans

Less confident on a phone. They check a message or call before they act.

GUARDIAN

Their children and grandchildren

Hear about a Stop, but only what the person chose to share.

ADMIN

Moderators

Approve community reports, and manage people, guides and areas.

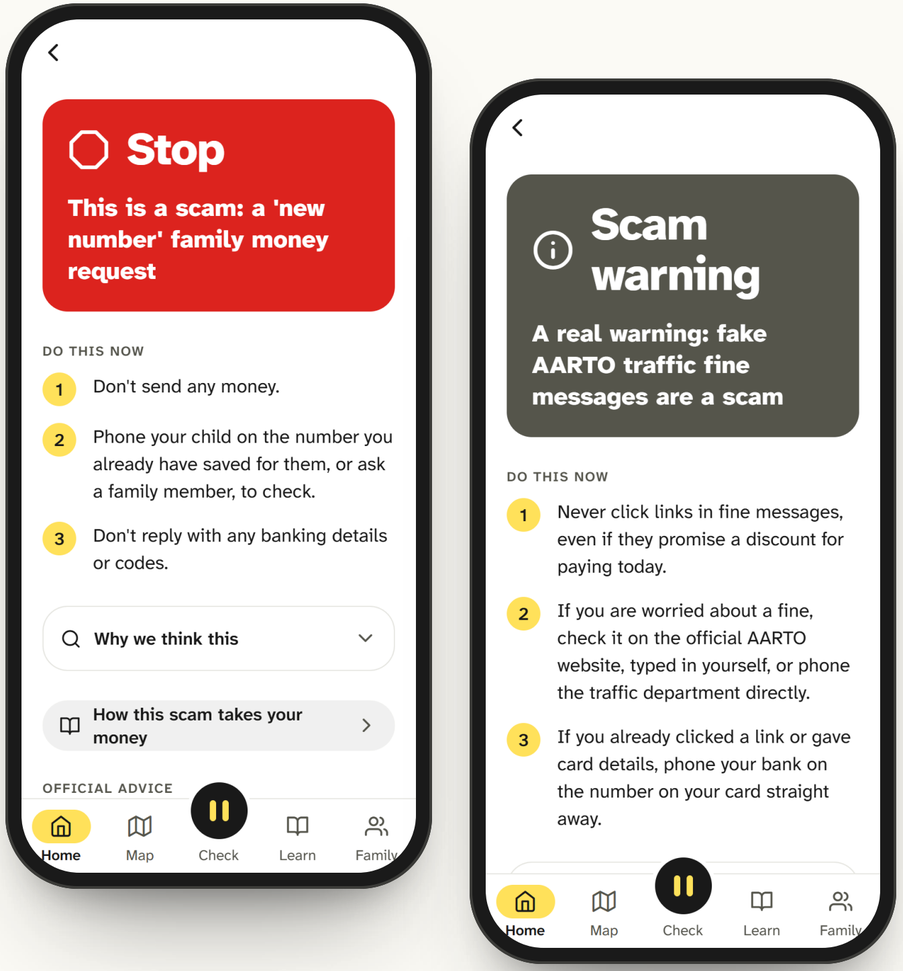
Atkinson Hyperlegible type **Big tap targets** **Plain words, no jargon** **Simple mode: one giant button**

THE ANSWER

One question: is this a scam?

1. **Paste, screenshot or speak it.** Calls too.
2. **Breathe** while it checks.
3. **A plain verdict**, and what to do now.
4. **Family hears about a Stop**, if you said so.

Stop **Wait** **Looks okay** **Scam warning**



DEMO VIDEO · RECORDED ON THE LIVE APP

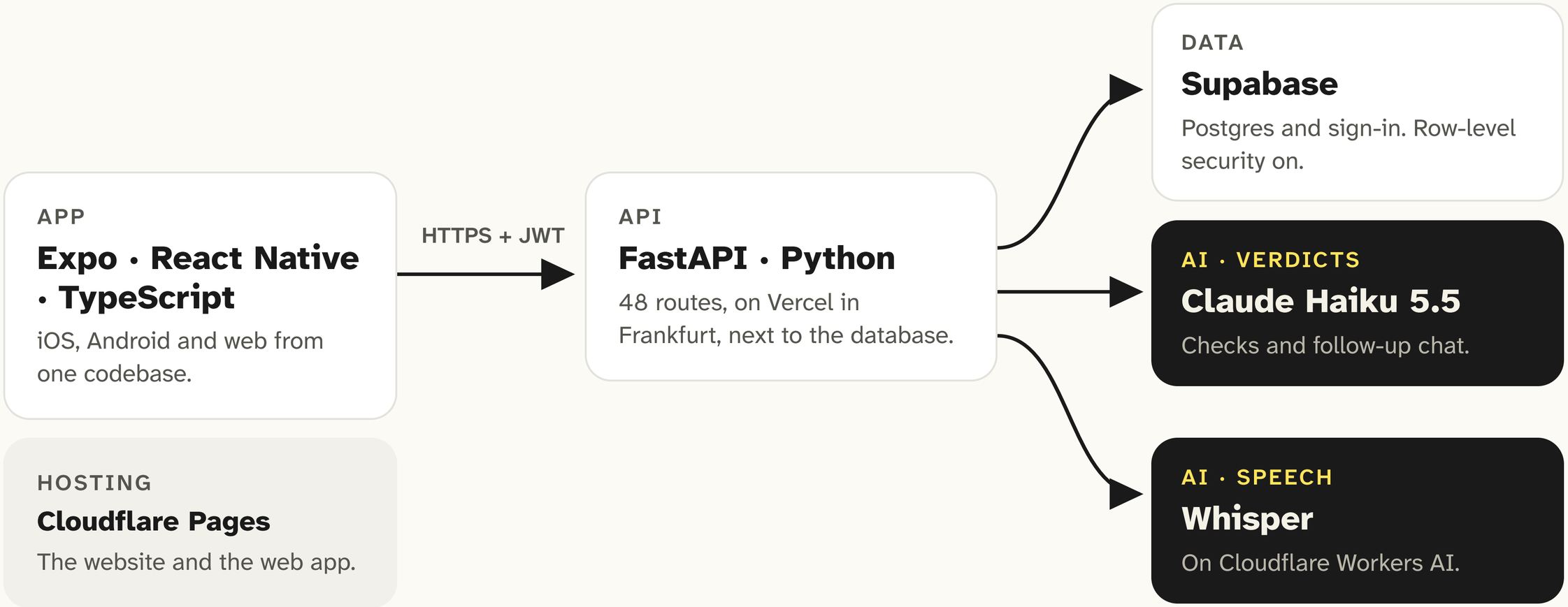
See it work.

1. The website, and the real app in the phone
2. Marie checks a “Hi Ma” WhatsApp: Stop
3. A screenshot of a fake SASSA SMS
4. A call in progress: “Say this, then hang up”
5. A warning poster gets a grey Scam warning
6. Report it, then the map and Learn
7. Pieter sees Marie’s alert
8. Admin: activity, moderation, people

Try it yourself at onhold.pages.dev: the demo accounts are on the sign-in screen.

HOW IT'S BUILT

One app, one API, two AI services.



THE AI IS THE CORE FEATURE

From a message to a verdict.

1. **Text, a screenshot, or speech**
   Whisper turns speech into words you can fix
2. **Claude reads it as untrusted data**
   SA scam patterns, and a prompt-injection guard
3. **Structured output**
   the reply always fits the schema on the right
4. **The server checks it**
   lengths, call scripts only for calls, source links
5. **Saved; family alerted on a Stop**
   only where sharing is switched on

```python
# backend/app/ai.py
class ScamVerdict(BaseModel):
    verdict: Literal["stop", "wait", "safe", "info"]
    confidence: float
    scam_type: ScamTypeLiteral | None = None
    headline: str = Field(min_length=1)
    reasons: list[str] = Field(min_length=1)
    next_steps: list[str] = Field(min_length=1)
    say_this: str | None = None
    is_call: bool
    sources: list[SourceId] = Field(default_factory=list)
```

SourceId is a fixed list of 8 official sites, so the AI can’t invent a link.

TESTED, NOT TRUSTED

29 known messages through the real model.

MODEL	RIGHT	DANGEROUS MISSES	FALSE ALARMS	SPEED	PER CHECK
Claude Sonnet 5 (before)	29 / 29	0	0	3.5 s	about R0.07
Claude Haiku 5.5 (now)	**29 / 29**	**0**	**0**	**3.0 s**	**about R0.005**

17

scams, including fake “official warnings” that push a link

10

genuine messages and real warnings that must not raise an alarm

~16×

cheaper after switching to Haiku, with nothing lost

“Dangerous” = a scam called Looks okay or Scam warning. Re-run with **uv run python -m scripts.eval**

TRUST AND SECURITY BY DESIGN

It shows its sources, and it can say it’s unsure.

Sources it can’t make up

Links come only from SABRIC, SASSA, SARS, AARTO, FSCA, PASA, SAPS.

Warnings are never green

A real warning poster is grey. A fake “warning” is a Stop.

Unsure means Wait

“On Hold can make mistakes. If something feels wrong, trust that.”

You choose what family sees

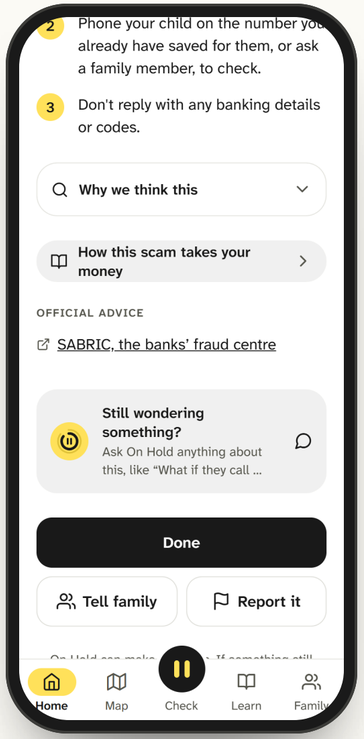
Per person, separate switches for verdicts and calls.

The API is the only door

Supabase sign-in tokens, row-level security, every query scoped to the person.

Guard rails

30 AI requests an hour per person; the last admin can’t be removed.



FULL CRUD, BY ROLE

Everything people make, they can manage.

WHAT	CREATE · READ · UPDATE · DELETE	WHO
Checks and follow-up chats	Create, read, delete	The person, own only
Scam reports	Create, read, edit, delete; approve or reject	Reporter, and admins
Family links	Invite, accept, change sharing, remove	Protected and guardian
People	Add, view history, edit name, area and role, delete	Admins
Learn guides	Create, read, edit, delete	Admins (everyone reads)
Map areas	Create, read, edit, delete	Admins

48 API routes, each documented in backend/docs/api.md

HOW IT WAS MANAGED

Small branches, tested on every push.

213
commits on main

38
short-lived feature, fix and docs branches

59
merges, each kept visible in the history

72
automated tests: 56 API, 16 app

42
bugs logged, each with its fixing commit

`feature/* fix/* docs/*` → `dev` → `main`

GitHub Actions on every push: pytest, TypeScript, lint and jest.

Website: Cloudflare Pages · App: Cloudflare Pages · API: Vercel · Data: Supabase · Speech: Workers AI

FEEDBACK, AND WHAT CHANGED

Testing changed the product.

FROM PEER USABILITY TESTING

[What testers said, 1]	→	*[What I changed]*
[What testers said, 2]	→	*[What I changed]*
[What testers said, 3]	→	*[What I changed]*

FROM TESTING THE LIVE APP

A real warning poster got a green “Looks okay”	→	**Grey Scam warning; fake warnings are Stop**
Speak returned the same canned words	→	**Real speech-to-text with Whisper**
Silence came back as the word “You”	→	**Voice detection: “We didn’t catch any words”**
The live app hung on database requests	→	**Transaction pooler; API moved next to the data**

All 42 bugs and fixes: docs/testing.md

WHAT'S NEXT

From reports to scam intelligence.

THE BIG ONE

Know what's going around before anyone reports it.

A daily AI job reads official alerts and drafts warnings with the source attached. An admin approves them first. In-app reports stay as the local, near-you layer.

SABRIC · SASSA · SARS · RTIA · FSCA → AI draft + source →

Admin approves → Learn and the map

App Store and Google Play

Build profiles are ready (Expo EAS).

Push alerts for family

Alerts are in-app today; push and SMS next.

Help during a call

Spotting a live scam call, Android first.

A bigger accuracy test

Built from real checks, with consent.

on hold

Thank you. Questions?

TRY IT

onhold.pages.dev

CODE

github.com/victordupreez0/On-Hold

DEMO SIGN-IN

marie@onhold.test · onhold-demo